AP CALCULUS AB

Unit 1 • Self Check 1.3 and 1.4 • Lessons 1-5 to 1-9

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Name: _____ Date: _____

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

Which function must have an absolute minimum on the interval?

(A) $\frac{1}{x}$ on $[1,3]$

(B) $\frac{1}{x}$ on $(0,3]$

(C) x on $(0,1)$

(D) $\ln x$ on $(0,1]$

Question 2

NO CALCULATOR

For $f(x) = x^2$, $\frac{f(1+h)-f(1)}{h} =$

(A) 2

(B) $2+h$

(C) $1+h$

(D) h

Question 3

NO CALCULATOR

For $g(x) = x^3$ on $[0,3]$, the MVT value is $c =$

(A) 1.5

(B) 3

(C) $\sqrt{3}$

(D) $\sqrt[3]{9}$

Question 4 (Free response)

NO CALCULATOR

Use the definition to find $f'(x)$ for $f(x) = x^2 - 3x$, write the tangent line at $x = 4$, and find the c given by the MVT on $[0,4]$.